

Solid Phase I of Carbon Dioxide: A New Gibbs Energy Equation of State and Its Application for the Calculation of Solid–Fluid Equilibria

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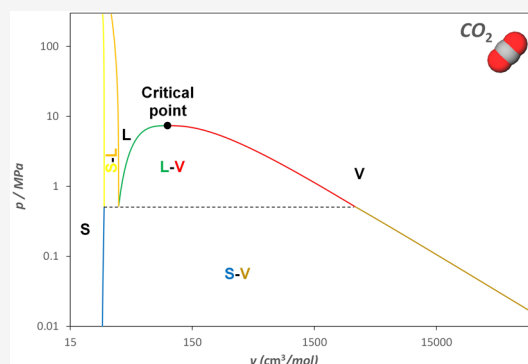
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ABSTRACT: A new Gibbs energy Equation of State (EoS) has been developed to describe solid phase I of carbon dioxide (“dry ice”) for use in phase equilibrium calculations. Its functional form, which uses temperature and pressure as independent variables, is inspired by the work of Stringari et al. (*J. Chem. Eng. Data* 2021, 66(2), 1157–1171) and is valid up to 400 K and 1.2 GPa. The model parameters were fitted to literature data for molar volume, isothermal compressibility, isobaric heat capacity, and the sublimation and melting pressures. Melting and sublimation lines were determined by coupling the new solid-phase EoS with the reference fluid-phase EoS of Span; Wagner (*J. Phys. Chem. Ref. Data* 1996, 25, 1509–1596). As illustrative applications, the new EoS was coupled with the default Helmholtz energy EoS implemented in REFPROP v10 to predict the solubility of CO₂ in methane and nitrogen. For comparison, similar calculations were performed using alternative solid-phase EoSs from the literature, namely those of (i) Jäger; Span (*J. Chem. Eng. Data* 2012, 57, 590–597), (ii) Trusler (*J. Phys. Chem. Ref. Data* 2011, 40(4), 043105), and the (iii) Classical Approach (Prausnitz et al. 1999).



1. INTRODUCTION

The main challenge with CO₂ is its high triple-point temperature compared to hydrocarbons and cryogenic fluids (methane, nitrogen, hydrogen, etc.). The deposition of CO₂ on the surface of heat exchangers found in cryogenic processes like natural gas/biomethane and hydrogen liquefactions can result in fouling, plant shutdowns, and other operational problems. Furthermore, Eggeman and Chafin pointed out that several commercial process simulators predict inaccurately CO₂ freeze conditions.¹ Another concern revolves around the inability of commercial heat exchangers models to account for phase changes, leading to inefficient process and plant designs.

Recently, Campestri et al.^{2,3} proposed a new modeling approach to predict solid–fluid equilibria of mixtures involving natural gas components, assuming that the heavy component in the mixture solidifies in its pure state. This approach works by coupling reference equations of state for the fluid and solid phases. A key advantage of this approach is that it does not require any parameter regression on solid–fluid equilibrium data, thereby making it a fully predictive approach.

The reference fluid-phase model is the reduced Helmholtz energy EoS based on the multifluid approximation, as implemented in REFPROP v10,⁴ whose functional form is equivalent to the GERG-2008 EoS.⁵ It is widely used as a reference model for fluid phase properties, developed from extensive experimental mixture data considering thermal and

caloric properties, and provides accurate predictions of fluid density, a key parameter for solubility calculations at solid–fluid equilibrium.

The reference solid-phase model is a Gibbs energy-explicit EoS, whose functional form was initially developed to describe the thermodynamic properties of solid methane and solid benzene.^{6,7}

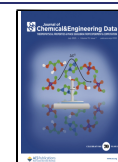
As part of an effort to develop a database of reference solid-phase models for various solids using a common functional form, and to extend and validate the modeling approach for predicting solid–fluid equilibria of mixtures involving these solids, the present work evaluates the applicability of the previously mentioned Gibbs energy-explicit EoS, developed for solid methane⁷ and solid benzene,⁶ to represent the thermodynamic properties of solid phase I of carbon dioxide. While it is acknowledged that various EoSs for solid phase I of carbon dioxide already exist in the literature,^{8,9} the primary aim of this study is to continue previous efforts to develop a solid-state EoS with a simple and easy-to-handle functional form, following the

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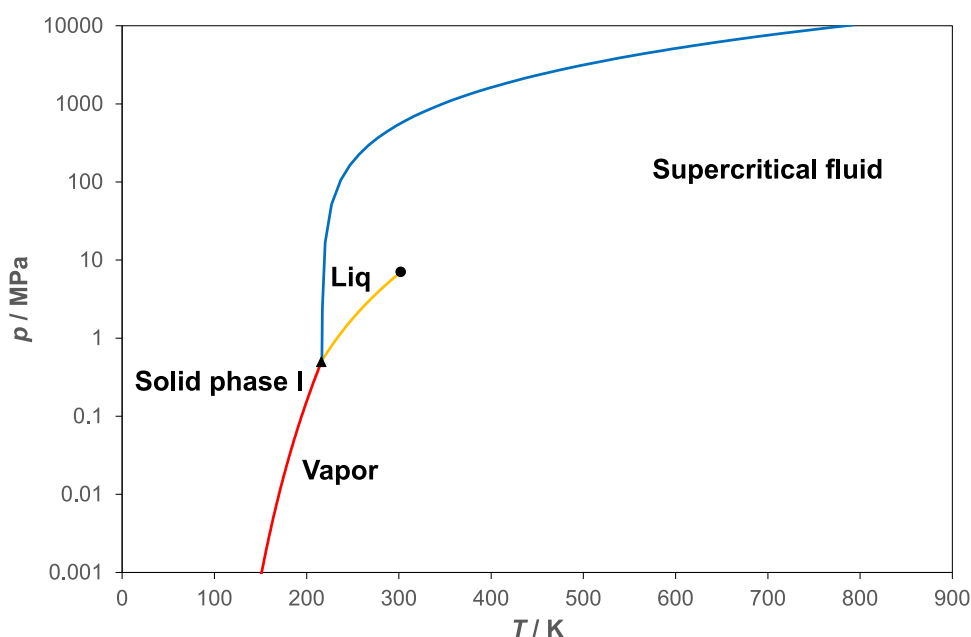


Figure 1. Phase diagram of CO₂ calculated using the correlation equations of Span and Wagner.¹⁰ ▲: triple point; ●: critical point; yellow dash: saturation line; blue dash: melting line; red dash: sublimation line.

Table 1. Literature Data for Phase I of Solid Carbon Dioxide

property	T range [K]	p range [MPa]	N data	primary (1)/secondary (2)	refs
isothermal compressibility, k	15–139	0.1	14	1	Manzhelii et al. ¹⁶
isobaric thermal expansion, α	13–139	0.1	20	2	Manzhelii et al. ¹⁶
	15–215	0.1	41	2	Krupskii et al. ¹⁷
molar volume, v	13–139	0.1	20	1	Manzhelii et al. ¹⁶
	20–114	0.1	7	2	Keesom and Köhler. ¹⁸
	20	0.1	1	1	Heuse ³⁶
	90–194	0.1	16	1	Maas and Barnes ¹⁹
	6–216	0.1	44	1	Krupskii et al. ¹⁷
	296	1000–10,000	19	1	Olinger ²⁰
	296	2410–12,000	17	1	Liu ¹⁴
	295–764	570–12,550	44	2	Giordano et al. ²¹
isobaric heat capacity, c_p	15–190	0.1	41	1	Giauque and Egan ²²
	2–38	0.1	60	1	Manzhelii et al. ¹⁶
	109–183	0.1	7	2	Maas and Barnes ¹⁹
	80–200	0.1	13	2	Eucken and Hauck ²³
	19–45	0.1	11	2	Eucken ²⁴
sublimation curve	154–196	1.51×10^{-3} – 1.11×10^{-1}	12	2	Giauque and Egan ²²
	106–154	1.33×10^{-7} – 1.343×10^{-3}	12	2	Tickner and Lossing ²⁵
	178–198	2.37×10^{-2} – 1.31×10^{-1}	16	1	Ambrose ²⁶
	201–213	1.80×10^{-1} – 3.84×10^{-1}	6	2	Baughman et al. ²⁷
	83–102	2.29×10^{-11} – 4.21×10^{-8}	20	2	Bryson et al. ²⁸
	154–217	1.50×10^{-3} – 5.26×10^{-1}	97	1	Bilkadi et al. ²⁹
	194–202	9.77×10^{-2} – 1.68×10^{-1}	7	1	Fernández-Fassnacht and del Rio ³⁰
	194–217	9.90×10^{-2} – 5.17×10^{-1}	22	2	Meyers and Van Dusen ³¹
melting curve	216–367	0.1–1177	12	2	Bridgmann ³²
	216–266	0.9–287	25	1	Michels et al. ³³
	216–222	0.5–24	20	1	Clusius et al. ³⁴
	300–391	531–1553	5	2	Grace and Kennedy ³⁵

methodology adopted in prior studies by Stringari et al.^{6,7} Particular focus is also placed on accurately calculating the phase equilibrium boundaries of CO₂ and accessing the model's capability to predict solid–fluid equilibria in selected binary mixtures involving CO₂ when coupled with the reference fluid-phase model.

Consequently, the first goal of this work is to develop a new Gibbs energy EoS for solid CO₂ (“dry ice”), similar to those previously published for solid methane⁷ and solid benzene,⁶ trying to obtain a proper description of the behavior of dry ice across the entire temperature and pressures ranges where experimental data exist. The modeling results obtained for solid

CO₂ in the single-phase region, as well as under melting and sublimation conditions (achieved by coupling the new EoS with the reference EoS for fluid CO₂¹⁰) have been compared to those obtained using two alternative literature EoSs for dry ice.^{9,11}

The secondary goal of this work is to compare predicted solid–fluid equilibria with experimental data for the nitrogen + carbon dioxide and methane + carbon dioxide mixtures. For this purpose, the default multiparametric EoS implemented in REFPROP v10⁴ was used to represent the fluid phases. Predictions were also performed by replacing the newly developed Gibbs EoS with the previously mentioned literature EoSs for dry ice.^{9,11} Furthermore, the Classical Approach¹² was also evaluated to highlight the advantages of using a Gibbs energy EoS over the simplest model commonly used in solid–fluid calculations.

1.1. Phase Diagram of Pure CO₂. Figure 1 illustrates the pressure–temperature phase equilibrium behavior of carbon dioxide, calculated using the correlation equations for melting, sublimation and vapor pressure proposed by Span and Wagner.¹⁰ The critical temperature and pressure of carbon dioxide are 304.1282 K and 7.3773 MPa, respectively, while the triple point temperature and pressure are 216.592 K and 0.51795 MPa.¹⁰

According to Santoro and Gorelli,¹³ solid phase I of carbon dioxide is a molecular solid in which the molecules interact solely by means of weak van der Waals forces. The oxygen atoms are bonded to the central carbon atom by two double bonds, giving the molecule a rigid linear structure. Due to its symmetric charge distribution, the molecules do not experience dipole–dipole interactions, leaving dispersion forces as the only intermolecular forces present.

Being part of the simplest molecular crystals, like N₂, dry ice exhibits a face-centered cubic structure with a *Pa*3 symmetry space group.¹³ As depicted in Figure 1, this phase remains stable at temperatures below 800 K and pressures up to approximately 10 GPa, although some studies reported stability up to 12.4 GPa.¹⁴ The different crystalline structures of CO₂ existing outside these temperature and pressure ranges are beyond the scope of this work and are not addressed.

Furthermore, quadrupole moments are considered significant for CO₂ molecules, with Trusler⁹ suggesting that they are responsible for the high triple and critical-point pressures of CO₂. Vega and Monson¹⁵ also pointed out that the presence of quadrupoles stabilizes solid structures that are not closely packed, which contributes to a reduction in the liquid range. This phenomenon helps explain the small ratio of the triple to critical-point temperatures observed in systems like CO₂ and acetylene.

1.2. Literature Data of Solid CO₂. Considerable attention was dedicated to collecting data for phase I of solid CO₂. Literature data on isothermal compressibility *k*, isobaric (cubic) thermal expansion α , molar volume on sublimation curve v_{sub} , isobaric heat capacity c_p , molar volume at high pressures v , sublimation and melting pressures, are summarized in Table 1. The temperature range, pressure range, and number of data points are indicated in the second, third, and fourth column of Table 1, respectively. For missing pressure data in the literature, the near-atmospheric pressure of 0.1 MPa was assigned. The impact of this assumption on the developed EoS was checked by testing different hypotheses for the pressure of these data, such as assigning the sublimation pressure from an independent correlation,¹⁰ with no significant changes observed in the regressed values of the EoS parameters. For each reference

indicated in the sixth column, the fifth column indicates whether the data were considered primary (1) or secondary (2). Primary data were used for the regression of the parameters of the EoS, while secondary data were used for comparison between model calculations and experimental results.

To the authors' knowledge, the only available data on the isothermal compressibility *k* for phase I of solid CO₂ is provided by Manzhelii et al.¹⁶ In their study, the isothermal compressibility was derived from in-house measurements of the speed of sound, molar volume, heat capacity, and thermal expansion, covering a temperature range of 15 to 139 K. As the experimental pressure is not reported in their article, it has been assumed to be 0.1 MPa for this work. Given the lack of alternative data sources, this data set has been considered as primary and used in the regression of the model's parameters.

In total, five sources of isobaric heat capacity data are available, ranging from 2 to 210 K.^{16,19,22–24} Manzhelii et al.¹⁶ were the only to perform c_p measurements at temperatures as low as 2 K and up to 38 K. Giauque and Egan²² covered a broader range, from 15 to 190 K. Given the similar quality of these two data sets, they were assigned as primary data.

However, accessing the reliability of c_p data for temperatures between 190 K and the triple point temperature is challenging. In this range, two sets of experimental data are available. The measurements by Eucken and Hauck²³ appear somewhat overestimated, while those of Maas and Barnes¹⁹ align with data of other researchers only at the lower end of the range. Consequently, the data sets of Eucken and Hauck,²³ Maas and Barnes¹⁹ and Eucken²⁴ were considered as secondary.

Molar volume of solid CO₂ was measured at elevated temperatures and pressures, covering a temperature range of 295 to 764 K and a pressure range of 490 MPa to 13 GPa.^{14,20,21,32} Using the X-ray technique, Olinger²⁰ investigated the molar volume across a pressure range of 1 to 100 kbar using a Bridgman anvil, while Liu¹⁴ measured molar volumes up to 50 GPa using a diamond-anvil cell, with both studies conducted at room temperature.

However, the high-pressure data of Liu¹⁴ may correspond to metastable solid states of either phase I or phase III, as the transition to phase III reportedly occurs with negligible change in molar volume. Since this work focuses solely on phase I of solid CO₂, only measurements below 13 GPa were considered. The primary data include measurements by Olinger,²⁰ Liu,¹⁴ and Giordano et al.²¹

Literature data on molar volumes along the sublimation curve range from 6 K up to the triple-point temperature. According to Trusler,⁹ the most accurate measurements come from the X-ray diffraction data of Krupskii et al.,¹⁷ covering a broad range of temperature, from 6 K up to 200 K with a relative uncertainty of 0.01%.¹⁷ The other three sources, Manzhelii et al.,¹⁶ Maas and Barnes,¹⁹ and Heuse³⁶ present similar qualities to Krupskii et al.¹⁷ Subsequently, all four data sets were also considered as primary data. The data provided by Keesom and Kohler¹⁸ were considered as secondary. As the authors did not provide the experimental sublimation pressures obtained in their work, the pressures were assumed to be 0.1 MPa for the regression of the model's parameters.

Eight literature sources reported the sublimation pressure of carbon dioxide: Meyers and Van Dusen,³¹ Giauque and Egan,²² Tickner and Lossing,²⁵ Ambrose,²⁶ Baughman et al.,²⁷ Bryson et al.,²⁸ Bilkadi et al.,²⁹ and Fernández-Fassnacht and del Rio.³⁰ These data sets extend from the triple point down to 60 K and pressures as low as 1.15×10^{-14} MPa. The sublimation data were

considered exceptionally good by Span and Wagner.¹⁰ Nevertheless, the data of Bilkadi et al.²⁹ had to be adjusted following the procedure outlined by Span and Wagner,¹⁰ as the thermometers were calibrated on an outdated triple-point temperature of CO₂.

Five literature sources examine the melting pressure of carbon dioxide: Bridgman,³² Michels et al.,³³ Clusius et al.,³⁴ Grace and Kennedy,³⁵ Giordano et al.³⁷ These data sets extend from the triple point up to 640 K and pressures as high as 4000 MPa. Similar to the case of Bilkadi et al.,²⁹ Span and Wagner¹⁰ identified inconsistencies between the data of Clusius et al.³⁴ and Michels et al.,³³ which required that their temperature values be corrected following the same procedure as previously mentioned.¹⁰ In addition, all available sublimation and melting data based on older temperature scales were converted to ITS-90 for consistency. The data of Giordano et al. was ultimately not considered as the authors only reported experimental data graphically.³⁷

In this work, the classification of sublimation and melting data as primary was guided by Span and Wagner's recommendations.¹⁰ Primary sublimation data include those from Ambrose,²⁶ Bilkadi et al.,²⁹ and Fernández-Fassnacht and del Rio,³⁰ while primary melting data include those from Michels et al.³³ and Clusius et al.³⁴ Each melting (or sublimation) pressure was calculated at the corresponding experimental temperature by equating the Gibbs energy of solid CO₂, as determined by the new EoS, to the Gibbs energy of liquid (or vapor) CO₂, as calculated using the reference EoS of Span and Wagner.¹⁰

The isobaric thermal expansion coefficient α of solid CO₂ was investigated by Manzhelii et al.¹⁶ and Krupskii et al.¹⁷ Manzhelii et al. argued that neither the X-ray nor pycnometer method provided satisfactory results for determining the thermal expansion coefficients. Instead, they used a quartz dilatometer method¹⁶ to investigate α between 13 and 139 K. Measurements at higher temperatures were deemed not feasible due to the high vapor pressure of dry ice.

In contrast, Krupskii et al. successfully extended the measurements to temperatures close to the triple point of CO₂, reaching as high as 215 K.¹⁷ Despite employing different techniques (quartz dilatometry and X-ray diffraction), both data sets are in agreement at temperatures below 210 K. However, noticeable deviations start to occur at temperatures above 210 K. In spite of the good quality of the data, the isobaric thermal expansion data were considered secondary in this work. This is due to the inability of the new EoS to describe the temperature dependence of the isobaric thermal expansion coefficient accurately, as further discussed in section 5.

1.3. Solid I–Liquid–Vapor Triple Point. The temperature, pressure, enthalpy, and Gibbs energy values at the solid I–liquid–vapor triple point of carbon dioxide were obtained using the reference Helmholtz EoS of Span and Wagner¹⁰ implemented in REFPROP v10.⁴ These values are reported in Table 2.

The enthalpy of transition between the liquid and solid I phases, reported as 8875 J mol^{−1} by Jäger and Span,¹¹ was selected for use in this work.

2. COMPUTATIONAL METHOD

The development of the Gibbs energy EoS for phase I of solid carbon dioxide follows the same analytical procedure developed by Stringari et al. for phase I of solid methane.⁷

In this work, eqs 1–3 were used to represent the isobaric heat capacity and the molar volume of phase I of solid carbon dioxide.

Table 2. Equation Parameters at the Vapor–Liquid–Solid I Triple Point

property	value	refs
temperature (K)	216.592	Lemmon et al. ⁴
pressure (kPa)	517.95	Lemmon et al. ⁴
enthalpy of liquid–solid I transition (J × mol ^{−1})	8875	Jäger and Span ¹¹
Gibbs energy of liquid and vapor (and solid I) (J × mol ^{−1})	−1447.0	Lemmon et al. ⁴
enthalpy of the liquid phase (J × mol ^{−1})	3522.3	Lemmon et al. ⁴

$$c_p^s(T \geq T_1) = G(1 + HT^{-1} + IT^{-2}) + LT^2 \quad (1)$$

$$c_p^s(T \leq T_1) = NT^3 + MT^6 \quad (2)$$

$$v^s(p, T) = A + F \frac{T}{1 + Ep} + J \left(\frac{T}{1 + Ep} \right)^2 + Bp + \frac{C}{D + p} + Z \ln(1 + Yp) \quad (3)$$

In eqs 1–3, T is the temperature in K, p is the pressure in MPa, A , B , C , D , E , F , G , H , I , J , L , M , N , Y , and Z are the parameters of the EoS. The temperature T_1 represents the point at which the isobaric heat capacity of dry ice switches from eqs 1 to 2, and vice versa, and was determined by regression to be 16.3 K. This is because the isobaric heat capacity of dry ice presents an inflection point near 30 K (see Figure 2), which renders the use of eq 1 (previously used to represent the heat capacities of solid methane⁷ and solid benzene⁶) insufficient to describe the entire temperature range covered by the available literature data. Consequently, a sixth-order polynomial (eq 2) was proposed in this work for temperatures below T_1 . It includes the Debye term (proportional to T^3) and a term proportional to T^6 due to deviations from the low-temperature Debye law.

Additionally, an extra term was added to the solid molar volume, namely the sixth term on the right-hand side of eq 3. This modification was made to better represent the molar volume of CO₂ at elevated pressures.

The regression of the parameters within eqs 1 and 2 has been done under the following constraints imposed at T_1 to ensure the continuity of the Gibbs energy at this temperature:

- Continuity of c_p^s , eq 4

$$(c_{peq\ 1a}^s)_{T=T_1} = (c_{peq\ 1b}^s)_{T=T_1} \quad (4)$$

- Continuity of the first derivative of c_p^s with temperature, eq 5

$$\left(\frac{\partial c_{peq\ 1a}^s}{\partial T} \right)_{T=T_1} = \left(\frac{\partial c_{peq\ 1b}^s}{\partial T} \right)_{T=T_1} \quad (5)$$

- Continuity of the second derivative of c_p^s with temperature, eq 6

$$\left(\frac{\partial^2 c_{peq\ 1a}^s}{\partial T^2} \right)_{T=T_1} = \left(\frac{\partial^2 c_{peq\ 1b}^s}{\partial T^2} \right)_{T=T_1} \quad (6)$$

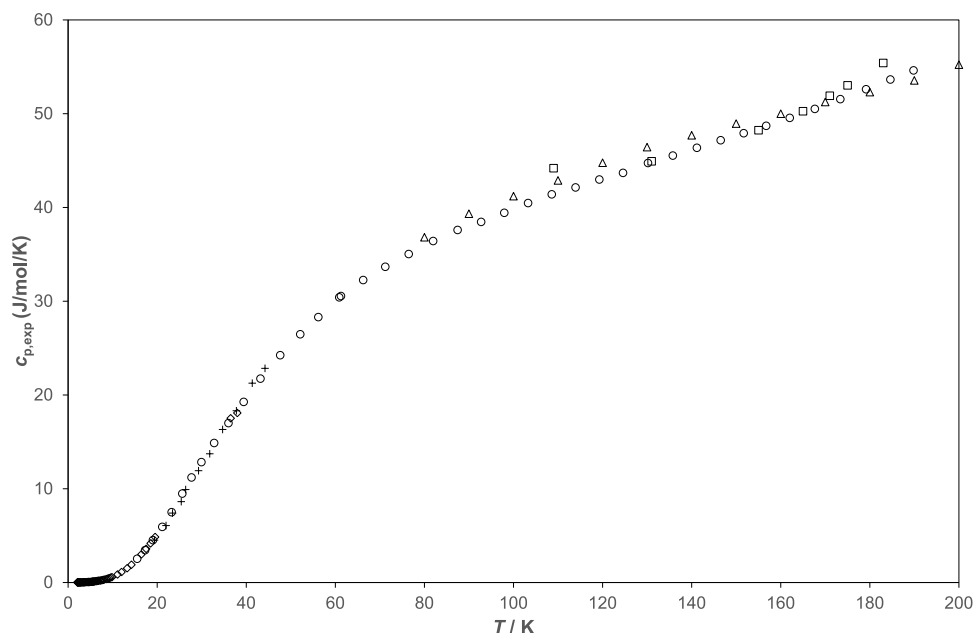


Figure 2. Experimental isobaric heat capacity of solid phase I of carbon dioxide versus temperature. $\diamond$: Manzhelii et al.;¹⁶ $\square$: Maas and Barnes;¹⁹ $\circ$: Giauque and Egan;²² $\triangle$: Eucken and Hauck;²³ $+$: Eucken²⁴

The analytical expressions of the first and second derivatives of eqs 1 and 2 with respect to temperature can be found in the [Supporting Information](#).

As previously demonstrated for solid methane⁷ and solid benzene,⁶ eqs 1–3 were derived based on two essential criteria:

1. A functional form that results in an analytical expression when integrated with respect to temperature and pressure.
2. A simple form with a minimum number of parameters to accurately represent the temperature and pressure dependence of the experimental data while maintaining, at the same time, a physically consistent behavior in temperature and pressure regions where data are scarce or absent.

After integrating eqs 1–3 and regrouping the terms with the same independent variables, the Gibbs energy of dry ice can be expressed as

$$\begin{aligned}
 g^s(p, T \leq T_1) &= A_0 + A_1 * T + A_2 * p^2 + A_3 * \ln(D + p) + A_4 \\
 &\quad * \frac{1}{1 + E * p} + A_5 * p + A_6 * \frac{p^2}{1 + E * p} + A_7 \\
 &\quad * \frac{T^2}{1 + E * p} + A_8 * \frac{T^2 * p}{1 + E * p} + A_9 * T \\
 &\quad * \ln(1 + E * p) + A_{14} * T^4 + A_{15} * T^7 + A_{16} \\
 &\quad * \ln(1 + Y * P) + A_{17} * P * \ln(1 + Y * P) \quad (7)
 \end{aligned}$$

$$\begin{aligned}
 g^s(p, T \geq T_1) &= A_0 + A_1 * T + A_2 * p^2 + A_3 * \ln(D + p) + A_4 \\
 &\quad * \frac{1}{1 + E * p} + A_5 * p + A_6 * \frac{p^2}{1 + E * p} + A_7 \\
 &\quad * \frac{T^2}{1 + E * p} + A_8 * \frac{T^2 * p}{1 + E * p} + A_9 * T \\
 &\quad * \ln(1 + E * p) + A_{10} * \ln(T) + A_{11} * T \\
 &\quad * \ln(T) + \frac{A_{12}}{T} + A_{13} * T^3 + A_{16} \\
 &\quad * \ln(1 + Y * P) + A_{17} * P * \ln(1 + Y * P) \quad (8)
 \end{aligned}$$

The relationships between the parameters A_i of eqs 7 and 8 and the parameters $A, B, C, D, E, F, G, H, I, J, L, M, N, Y$, and Z of eqs 1–3 are provided in the [Supporting Information](#).

The regressed values of the parameters of eqs 7 and 8 are presented in [Table 3](#).

Having established the mathematical expression for the Gibbs energy of the solid phase I of carbon dioxide, the isothermal compressibility k , isobaric thermal expansion α , molar volume v , isobaric heat capacity c_p can be determined using eqs 9–12:

$$k^s(p, T) = -\frac{1}{v^s} \left(\frac{\partial v^s}{\partial p} \right)_T = -\frac{g_{pp}^s}{g_p^s} \quad (9)$$

$$\alpha^s(p, T) = \frac{1}{v^s} \left(\frac{\partial v^s}{\partial T} \right)_p = \frac{g_{pT}^s}{g_p^s} \quad (10)$$

$$v^s(p, T) = g_p^s \quad (11)$$

$$c_p^s(p, T) = -T g_{TT}^s \quad (12)$$

Table 3. Parameters of Equations 7 and 8

parameter	value eq 7	value eq 8	units
p_0	0.1		MPa
T_0	13		K
A_0	-1.091966×10^4	-1.357561×10^4	$\text{J} \times \text{mol}^{-1}$
A_1	3.5128018×10^2	9.242542×10^1	$\text{J} \times \text{mol}^{-1} \times \text{K}^{-1}$
A_2	0		$\text{J} \times \text{mol}^{-1} \times \text{MPa}^{-2}$
A_3	4.887214×10^1		$\text{J} \times \text{mol}^{-1}$
A_4	-2.559126		$\text{J} \times \text{mol}^{-1}$
A_5	2.886850×10^1		$\text{J} \times \text{mol}^{-1} \times \text{MPa}^{-1}$
A_6	$1.675765 \cdot 10^{-6}$		$\text{J} \times \text{mol}^{-1} \times \text{MPa}^{-2}$
A_7	$-4.630560 \cdot 10^{-6}$		$\text{J} \times \text{mol}^{-1} \times \text{K}^{-2}$
A_8	$4.630560 \cdot 10^{-5}$		$\text{J} \times \text{mol}^{-1} \times \text{MPa}^{-1} \times \text{K}^{-2}$
A_9	$2.693343 \cdot 10^{-1}$		$\text{J} \times \text{mol}^{-1} \times \text{K}^{-1}$
A_{10}		-1.516896×10^3	$\text{J} \times \text{mol}^{-1}$
A_{11}		-4.989449×10^1	$\text{J} \times \text{mol}^{-1} \times \text{K}^{-1}$
A_{12}		-6.106386×10^3	$\text{J} \times \text{K} \times \text{mol}^{-1}$
A_{13}		$-5.795455 \cdot 10^{-5}$	$\text{J} \times \text{mol}^{-1} \times \text{K}^{-3}$
A_{14}	$-4.686085 \cdot 10^{-5}$		$\text{J} \times \text{mol}^{-1} \times \text{K}^{-4}$
A_{15}	$-5.671194 \cdot 10^{-10}$		$\text{J} \times \text{mol}^{-1} \times \text{K}^{-7}$
A_{16}	-4.529889×10^3		$\text{J} \times \text{mol}^{-1}$
A_{17}	-3.277236		$\text{J} \times \text{mol}^{-1} \times \text{MPa}^{-1}$
D	2.508821×10^2		MPa
E	$8.092092 \cdot 10^{-4}$		MPa^{-1}
Y	$7.234695 \cdot 10^{-4}$		MPa^{-1}

where $g_T^s = \left[\frac{\partial g^s(p, T)}{\partial T} \right]_p$, $g_p^s = \left[\frac{\partial g^s(p, T)}{\partial p} \right]_T$, $g_{TT}^s = \left[\frac{\partial^2 g^s(p, T)}{\partial T^2} \right]_p$,
 $g_{pp}^s = \left[\frac{\partial^2 g^s(p, T)}{\partial p^2} \right]_T$, and $g_{pT}^s = \left[\frac{\partial^2 g^s(p, T)}{\partial p \partial T} \right]$.

Expressions for g_T^s , g_p^s , g_{TT}^s , g_{pp}^s , and g_{pT}^s are provided in the Supporting Information. Due to the simple functional forms of eqs 7 and 8, and the direct calculation of all thermodynamic properties without requiring iterative procedures, the model has been implemented in a Microsoft Excel file. For solid–liquid and

solid–vapor equilibria calculations, the coupling of eqs 7 and 8 with the reference fluid model of Span and Wagner¹⁰ was implemented in the Microsoft Excel version of REFPROP v10.⁴

The parameters in eqs 7 and 8 were regressed using the primary data provided in Table 1, with the corresponding values for each property calculated via eqs 9–12. The objective function, OF, used for the regression of the parameters was the sum of the absolute average deviation, (AAD)%, relative to the primary data for isothermal compressibility, molar volume, isobaric heat capacity, sublimation and melting pressures, eq 13.

$$\text{OF} = \frac{\sum_{i=1}^{N_p} M_i \text{AAD}\%_i}{\sum_{i=1}^{N_p} M_i} \quad (13)$$

where N_p is the number of thermodynamic properties considered, M_i is the number of primary data for property i , and $\text{AAD}\%_i$ is the absolute average deviation (AAD%) calculated with respect to the primary data for property i .

The parameters of eqs 1 and 2 were initially regressed using primary data for isobaric heat capacity data to obtain their initial values. Subsequently, using these initial values for eqs 1 and 2, the parameters of eqs 7 and 8 were then simultaneously regressed on the primary data for molar volume, isothermal compressibility, isobaric heat capacity, sublimation and melting pressures.

3. RESULTS AND DISCUSSION

3.1. Comparison of the New EoS with Experimental Data of Solid CO₂. The comparison between the model and the experimental data was evaluated using statistical indices, namely Absolute Average Deviation (AAD), Bias, and Maximum Absolute Deviation (MAD), with their expressions provided in the Supporting Information.

The agreement between eqs 7 and 8 and the experimental data is illustrated in Figure 3 (molar volume at low pressures), Figure 4 (isothermal compressibility coefficient), Figure 5 (thermal expansion coefficient), Figure 6 (isobaric heat capacity), Figure

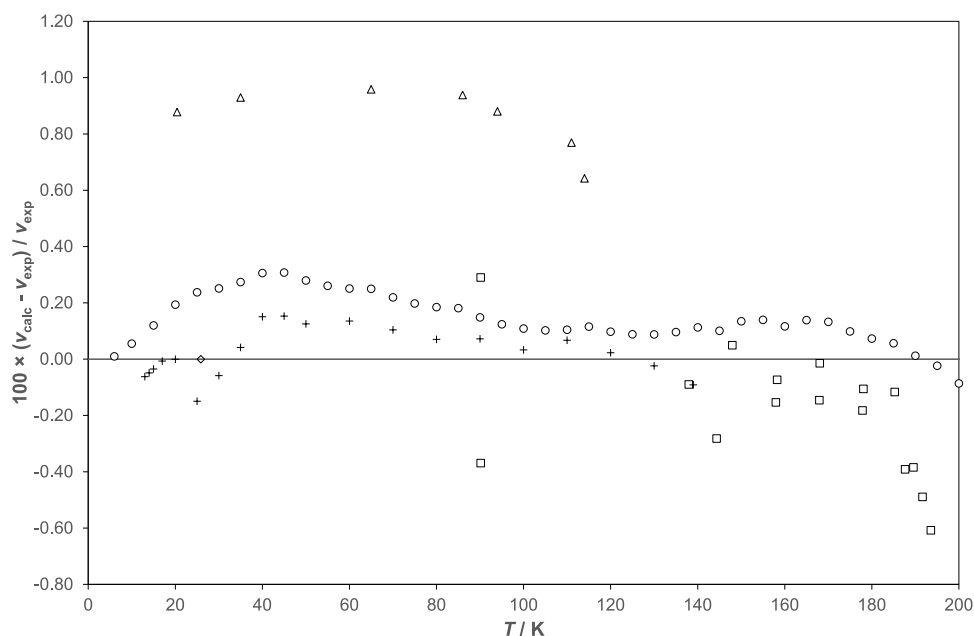


Figure 3. Deviation between calculated and experimental solid CO₂ molar volumes at low pressure. + : Manzhelii et al.;¹⁶ O : Krupskii et al.;¹⁷ Δ : Keesom and Köhler;¹⁸ ◇ : Heuse;³⁶ □ : Maas and Barnes¹⁹

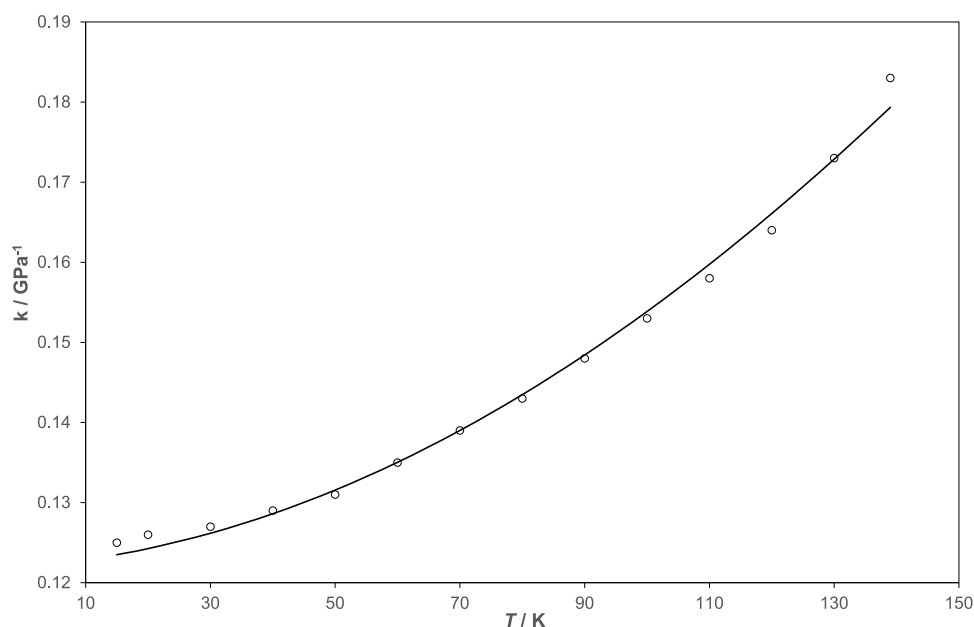


Figure 4. Calculated and experimental isothermal compressibility coefficients of solid CO₂. O: Manzhelii et al.;¹⁶ —: eqs 6 and 7

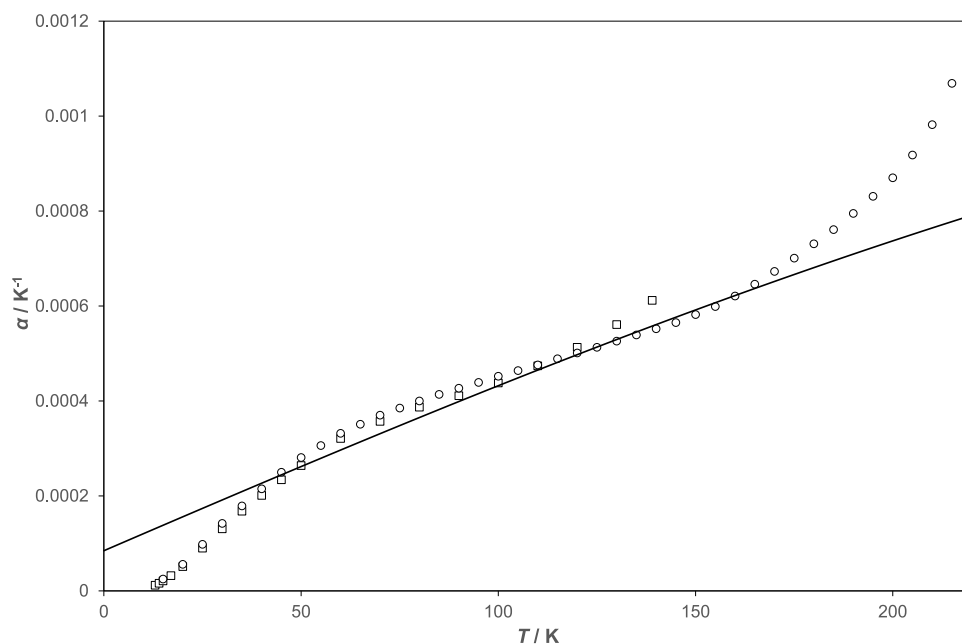


Figure 5. Calculated and experimental thermal expansion coefficients of solid CO₂. O: Krupskii et al.;¹⁷ □: Manzhelii et al.;¹⁶ —: eqs 6 and 7

7 (molar volume at high pressures), Figures 8 and 9 (sublimation pressure), and Figure 10 (melting pressure).

The new EoS represents the primary data for molar volume with an AAD of 0.15%, a near-zero Bias of 0.022% and a MAD of 0.61%. The low Bias relative to the AAD suggests that the model does not exhibit a systematic shift with respect to the data. The secondary data from Keesom and Köhler¹⁸ are represented with an ADD of 0.86%, and a MAD of 0.96%. Given that the data of Keesom and Köhler¹⁸ deviate more significantly from the others, this may indicate a systematic error in their X-ray measurements of the molar volume. Despite the higher deviations for the data of Keesom and Köhler¹⁸ compared to the primary data, Figure 3 shows that the molar volumes calculated using the new EoS are in very good agreement with the experimental data at low pressures.

As shown in Figure 4, which depicts the plot of isothermal compressibility data and the values calculated from eqs 7 and 8, the new EoS provides satisfactory results. It represents the data of Manzhelii et al.¹⁶ with an AAD of 0.69%, a Bias of −0.12%, and a MAD of 2.0% at approximately 140 K.

The thermal expansion coefficient data and values obtained from eqs 7 and 8 are plotted against temperature in Figure 5. From the figure, it can be observed that the new EoS predicts a linear temperature dependence of α in the investigated temperature range, in contrast to the polynomial behavior observed in the data of Krupskii et al.¹⁷ and Manzhelii et al.¹⁶ As a consequence, MAD as high as 1000% was observed.

Since α depends on the first partial derivative of the solid molar volume with respect to temperature at constant pressure (see eq 10), the expression for the solid molar volume proposed

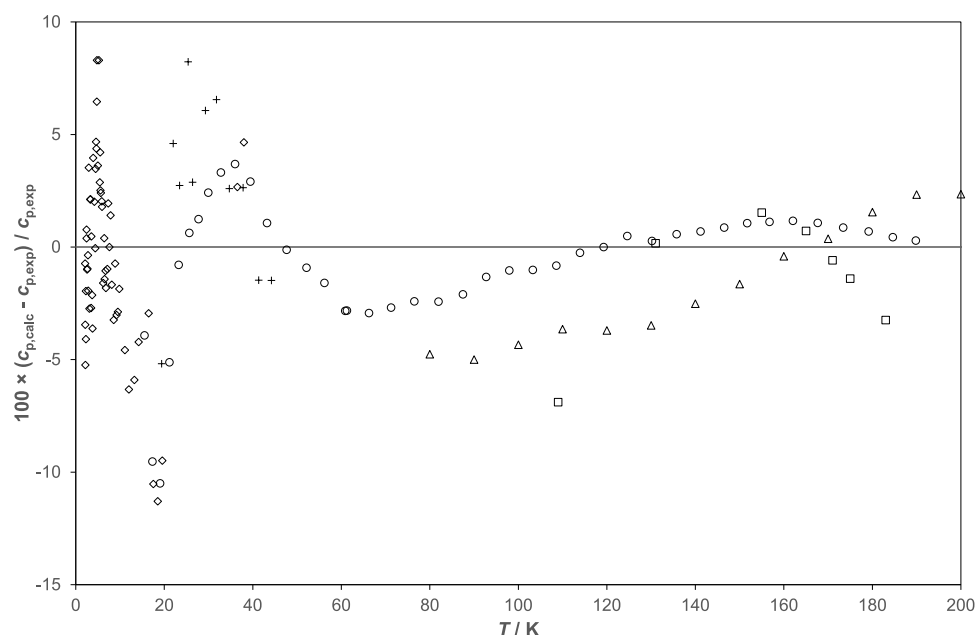


Figure 6. Deviation between calculated and experimental isobaric molar heat capacities of solid CO₂. ◇: Manzhelii et al.;¹⁶ □: Maas and Barnes;¹⁹ ○: Giauque and Egan;²² △: Eucken and Hauck;²³ + : Eucken²⁴

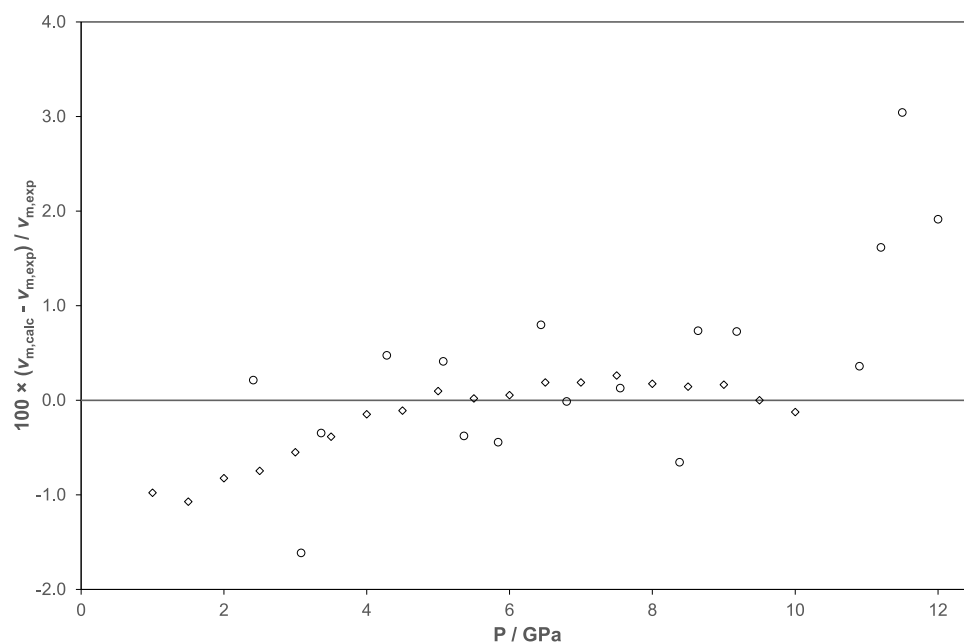


Figure 7. Deviation between calculated and experimental isothermal high-pressure molar volumes of solid CO₂ at $T = 296$ K. ○: Liu;¹⁴ ◇: Olinger²⁰

in this work (eq 3) – whose temperature dependence is similar to that of solid benzene⁶ and solid methane⁷ (molecules whose α shows a linear temperature evolution trend) – results in a first-order polynomial in temperature (eq 14), leading to a linear variation of the thermal expansion coefficient with temperature at a given pressure.

$$\left(\frac{\partial v_s}{\partial T}\right)_p = \frac{F}{1 + Ep} + \frac{2J}{(1 + Ep)^2}T \quad (14)$$

That being said, data from Krupskii et al.¹⁷ suggest that the thermal expansion coefficient of dry ice, along with its heat capacity, exhibits a cubic temperature dependence, which further explains why the new EoS fails to reproduce this

behavior. A higher order of temperature dependence, which the authors determined to be approximately $v(T) \propto T^5$, could resolve this issue. However, this would require adding new coefficients to eqs 7 and 8, rendering the equation more complex without providing significant advantages in the representation of phase equilibria. It was found that this property has minimal effect on the calculation of the Gibbs energy of carbon dioxide in the solid I phase.

Figure 6 shows the percentage deviations between the experimental isobaric heat capacity and the values calculated using the new EoS (eqs 7 and 8). For the primary data considered, the new EoS gives an overall AAD of 2.65%, a Bias of −0.55%, and a MAD of 11.29%. As for the secondary data, the new EoS gives an AAD of 3.07%, a Bias of −0.14%, and a MAD

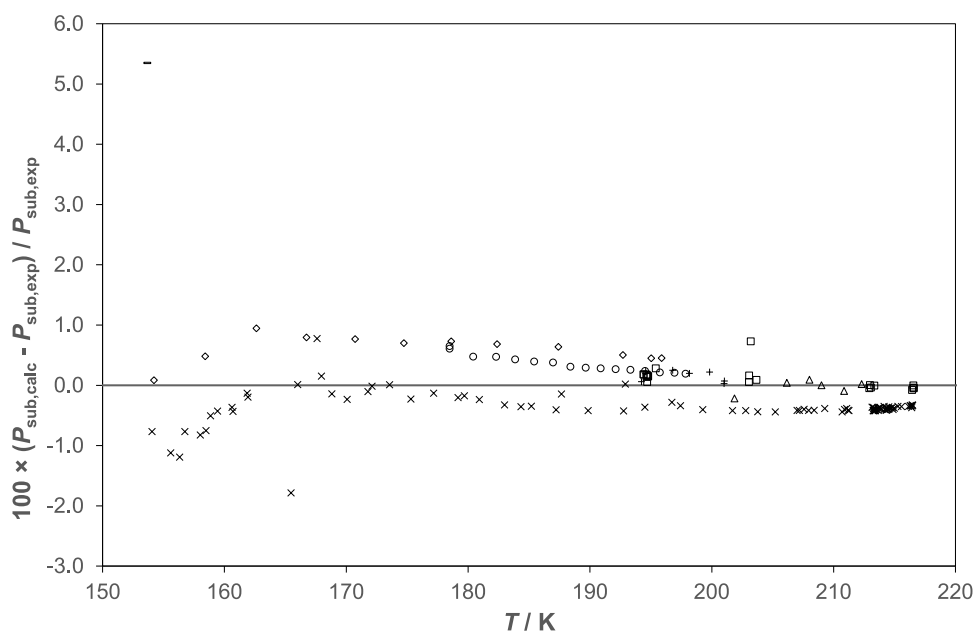


Figure 8. Deviation between calculated and experimental sublimation pressures of CO₂ for $T > 150$ K. $\diamond$: Giauque and Egan;²² —: Tickner and Lossing;²⁵ $\circ$: Ambrose;²⁶ $\triangle$: Baughman et al.;²⁷ $\times$: Bilkadi et al.;²⁹ $+$: Fernández-Fassnacht and del Rio;³⁰ $\square$: Meyers and Van Deusen³¹

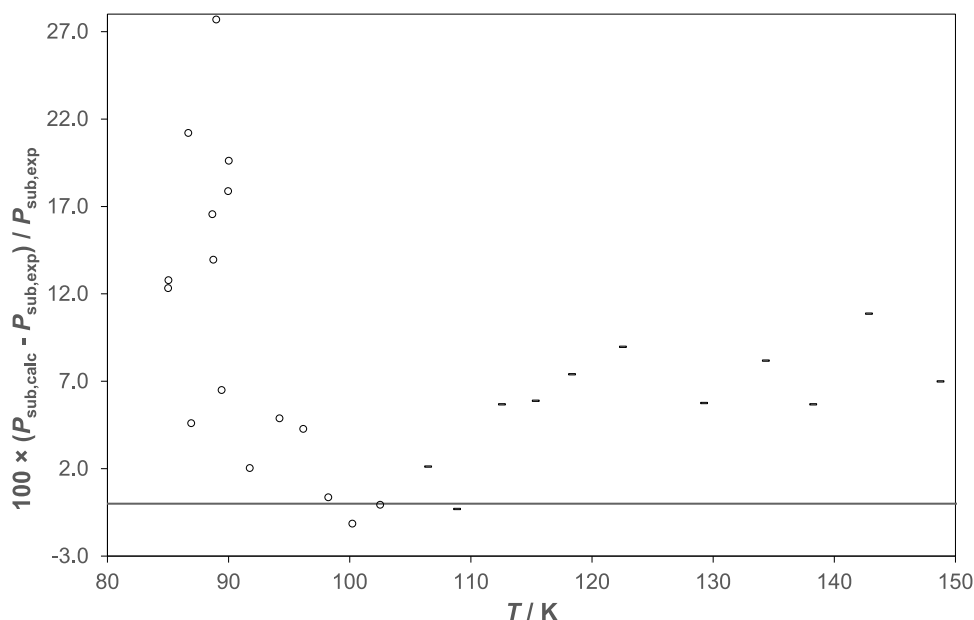


Figure 9. Deviation between calculated and experimental sublimation pressures of CO₂ for $T < 150$ K. —: Tickner and Lossing;²⁵ $\circ$: Bryson et al.²⁸

of 8.23%. The high MAD observed is a result of high deviations around T_1 . This deviation reveals the slight inability of the proposed model to accurately describe the rapid change in heat capacity around this temperature despite using two functions (eqs 1 and 2).

Figure 7 shows the percentage deviations between the experimental values of high-pressure molar volume data at 296 K and the modeling results. Since the transition to phase III of solid carbon dioxide is believed to occur around 12 GPa, the data of Liu¹⁴ exceeding this pressure were not considered in the quantitative comparison. The new EoS is able to cover a wide range of pressures and temperatures, accurately describing the molar volume of dry ice at high temperatures and high pressures. For all the primary data considered, eqs 6 and 7 yield an AAD of 0.74%, a Bias of -0.38% , and a MAD of 3.04%.

Figure 8 illustrates the comparison between experimental and calculated sublimation pressures at temperatures above 150 K. The primary data chosen for sublimation pressures fall within this temperature range, and the new Gibbs energy EoS gives an AAD of 0.38%, a Bias of -0.25% , and a MAD of 1.55%.

Two literature sources, namely Tickner and Lossing²⁵ and Bryson et al.,²⁸ reported sublimation pressures for temperatures below 150 K. As these data sets were considered secondary in this work, the calculation of sublimation pressure in this region is predictive. The comparison between calculated and experimental sublimation pressures extends down to $2.293 \cdot 10^{-11}$ MPa at 82.96 K (data from Bryson et al.²⁸). The comparison shown in Figure 9 indicates that the vapor pressures of Tickner and Lossing²⁵ and Bryson et al.²⁸ are systematically lower than the predicted values, except for three measurements. Despite the

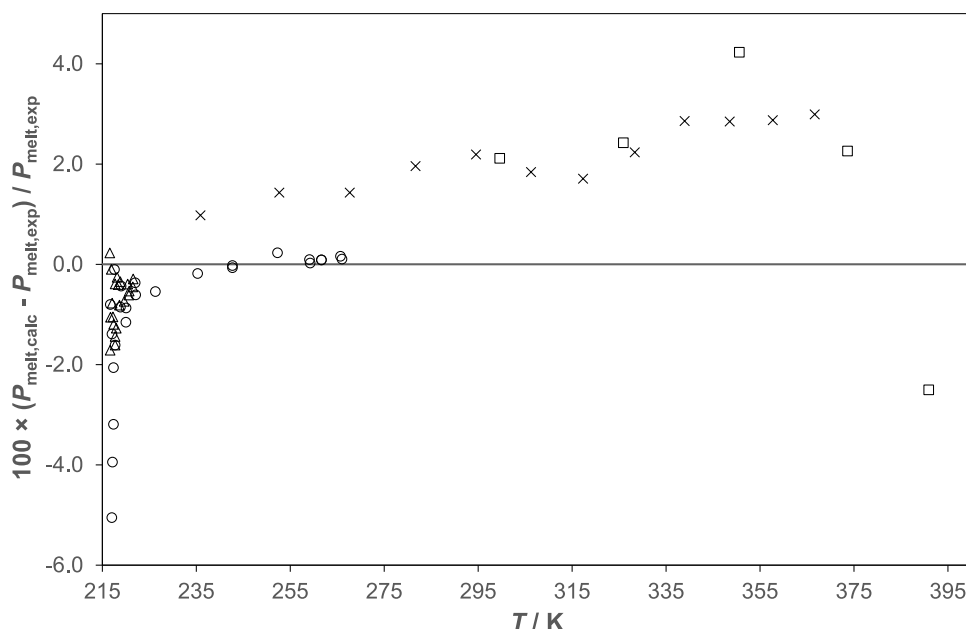


Figure 10. Deviation between calculated and experimental melting pressures of CO₂. × : Bridgman;³² ○ : Michels et al.;³³ △ : Clusius et al.;³⁴ □ : Grace and Kennedy³⁵

apparent high percentage deviations between the calculations and the data in Figure 9, the authors consider that the model shows satisfactory agreement with the secondary data.

Figure 10 compares the experimental melting pressures with the values predicted by coupling the reference fluid-phase EoS of carbon dioxide¹⁰ with the new EoS of solid carbon dioxide at temperatures up to 400 K. The model shows excellent agreement with the primary data of Clusius et al.³⁴ and Michels et al.,³³ achieving an overall AAD of 0.88%, Bias of −0.84% and MAD of 3.57%.

Although the reference fluid-phase EoS of Span and Wagner¹⁰ is only valid up to pressures of 800 MPa, the comparison between the experimental data and the calculated values was extended to pressures as high as 1600 MPa at $T = 400$ K, twice the maximum validity limit of the fluid-phase EoS. As shown in Figure 10, the predictions remain relatively satisfactory in comparison with the secondary data, yielding an AAD of 2.29%, Bias of 1.99% and MAD of 5.05%.

3.2. Comparison of the New EoS with Literature EoSs for Solid CO₂. This section presents the comparison between the model proposed in this work (called Model A) and two literature models for solid carbon dioxide (called Model B and Model C).

This work adopted a thermodynamic path similar to the model of Jager and Span¹¹ to describe the Gibbs energy EoS for solid CO₂ (Model B, eq 15).

$$g(p, T) = h_0 - Ts_0 + \int_{T_0}^T c_p(T, p_0) dT - T \int_{T_0}^T \frac{c_p(T, p_0)}{T} dT + \int_{p_0}^p v(p, T) dp \quad (15)$$

$$c_p(T, p_0) = \frac{aT^5 + bT^3}{T^4 + cT^2 + d} \quad (16)$$

The functional form of the isobaric heat capacity along the isobar p_0 (eq 16) is inspired by Feistel and Wagner,³⁸ who proposed a similar expression using a Padé approximation to

combine Debye's cubic law at low temperatures with the linear behavior at high temperatures.

Instead of proposing an explicit expression for the molar volume, Jager and Span¹¹ described the partial derivatives of the molar volume of dry ice with respect to pressure and temperature, as expressed in eq 17.

$$v - v_0 = \int_{v_0}^v dv = \int_{T_0}^T \frac{\partial v}{\partial T}(T, p_0) dT + \int_{p_0}^p \frac{\partial v}{\partial p}(T, p) dp \quad (17)$$

As a result, the final expression for the Gibbs energy EoS of solid CO₂ is

$$\begin{aligned} \frac{g}{RT_0} = & g_0 + g_1 \Delta v + g_2 \Delta v^2 + g_3 \left\{ \ln \left(\frac{v^2 + g_4^2}{1 + g_4^2} \right) \right. \\ & \left. - \frac{2v}{g_4} \left[\arctan \left(\frac{v}{g_4} \right) - \arctan \left(\frac{1}{g_4} \right) \right] \right\} \\ & + g_5 \left\{ \ln \left(\frac{v^2 + g_6^2}{1 + g_6^2} \right) - \frac{2v}{g_6} \left[\arctan \left(\frac{v}{g_6} \right) \right. \right. \\ & \left. \left. - \arctan \left(\frac{1}{g_6} \right) \right] \right\} + g_7 \Delta \pi [e^{f_a(v)} + K(v)g_8] \\ & + g_9 K(v) [\pi + g_{10}]^{n-1/n} - (1 + g_{10})^{n-1/n} \quad (18) \end{aligned}$$

Readers are encouraged to refer to the work of Jager and Span¹¹ for a detailed derivation of eq 18 and the interpretation of its corresponding terms.

Trusler⁹ proposed an expression for the molar Helmholtz energy of inorganic crystalline solids (Model C, eq 19) using the Quasi-Harmonic Approximation (QHA), a method previously used by Giordano et al. to determine the thermal pressure term found in the Mie–Grüneisen EoS.²⁰ In Trusler's approach, the density of states was described using the Debye–Einstein

Table 4. Performance of the Three Models in Terms of AAD%, Bias%, MAD% in the Calculation of the Primary and Secondary Data Chosen in This Work^a

Property	Model A (this work)			Model B (Jager and Span ¹¹)			Model C (Trusler ⁹)		
	AAD%	Bias%	MAD%	AAD%	Bias%	MAD%	AAD%	Bias%	MAD%
ν^S (SVE)	0.21	0.089	0.96	0.23	0.18	1.01	0.15	-0.02	0.79
c_p^S	2.75	-0.45	11.29	26.91	25.92	80.15	1.74	-0.48	7.11
k	0.69	-0.12	2.00	1.29	-0.16	2.25	8.08	-8.08	11.37
α	66.70	57.63	991.73	4.97	-2.18	24.41	4.91	3.67	40.81
p SVE ($T > 80$ K)	1.99	1.58	38.60	2.40	2.04	41.86	1.93	1.38	37.78
p SLE ($T < 400$ K)	1.26	-0.06	5.05	1.09	-0.59	12.91	0.83	0.29	3.97
ν^S (296 K)	0.56	0.09	3.04	0.90	-0.10	3.90	0.54	-0.06	1.20
ν^S (high T&P)	0.89	-0.76	2.56	29.25	-29.13	80.81	0.56	-0.08	1.70

^aGreen color has been used to point out the lowest deviations for each property.

approximation. Specifically, the Debye distribution accounts for the intramolecular vibrational modes (vibrons), while the sum of discrete harmonic terms (Einstein) represents the rotational vibrational modes (librons) of the molecules within the lattice.

$$A_m(T, V_m) = A_0(V_m) + A_D(T, V_m) + \sum_{i=1}^m a_i A_i(T, V_m) + A_{int}(T) + A_A \quad (19)$$

The terms in eq 19 are as follows: A_0 under the extreme condition of $T = 0$ K, accounting for the intermolecular and zero-point lattice energies of solid CO₂ crystal; A_D from the Debye distribution of lattice vibrations; A_i from the i th discrete harmonic component of the lattice librations (five terms in total); A_{int} from the internal molecular vibrations; last, A_A to take into account the deviations from the QHA, seen as an anharmonic correction term.

Other thermodynamic properties of the system may be derived by differentiating the various terms of eq 19.

Table 4 summarizes the quantitative comparison (in terms of AAD%, Bias%, MAD%) between experimental primary and secondary data selected in this work and values generated by the three models.

With respect to the molar volume along the sublimation curve, ν^S (SVE), all three models yield comparable AAD% results, with model C returning the lowest Bias. Model B slightly underestimates the molar volume, giving a positive bias of 0.18%. The MAD% for all three models is 1% or less, indicating a high level of accuracy.

Model C performs best when compared to experimental isobaric heat capacity (c_p^S) data, followed by Model A, and last Model B. Below 50 K, the AAD% of Model B increases as the temperature decreases, peaking at 80% around 5 K. This behavior highlights the limitation of Model B to describe the isobaric heat capacity accurately at extremely low temperatures, which is unsurprising given that the model's valid temperature range starts above 80 K.

Model A provides the most accurate representation of the experimental isothermal compressibility (k) data, followed by

Model B. In contrast, Model C overestimates the values. Although the MAD% of Model C appears seemingly high at 11%, it is relatively small when compared to the magnitude of the isothermal compressibility of CO₂ (on the order of 10⁻³ GPa). Thus, an error of 10% corresponds to an absolute difference of 10⁻⁴ GPa.

As explained previously, the model under study struggles to describe the temperature variation of the isobaric thermal expansion (α) accurately, as reflected by a MAD exceeding 100%. This limitation arises because the derivative of molar volume over pressure yields a first-order temperature dependency, whereas a higher-order dependency is required to well describe the experimental evolution of k with temperature. In contrast, Models B and C provide satisfying results, with a high MAD% attributed to the results of Manzhelii et al.¹⁶

All three models achieved excellent results in the determination of the sublimation pressure, p SVE ($T > 80$ K), with AADs % and Bias% around 2%. Among the models, Model C provides the best performance across all given statistical indexes.

All three models perform similarly in the calculation of the melting pressure, p SLE ($T < 400$ K), with Model C achieving an AAD below 1%. Model A and B slightly underestimates the melting pressure, whereas Model C overestimates it.

Models A and C perform well for molar volumes at 296 K, ν^S (296 K), and under high temperature and pressure conditions, ν^S (high T&P). Given that Model B is valid only between 80 and 300 K, it comes as no surprise that the model exhibits significant deviations outside this temperature range.

Considering all the thermodynamic properties studied in this work, Model C performs the best in terms of statistical indices. Although sophisticated, the model accurately describes the properties of solid phase I of carbon dioxide with minimal errors and has a broad validity range covering temperatures up to 800 K and pressures up to 12 GPa. However, its reliance on the Debye function to evaluate the contribution of the harmonic oscillators to the total Helmholtz energy requires the use of numerical integration methods, rendering the model slightly challenging to implement.

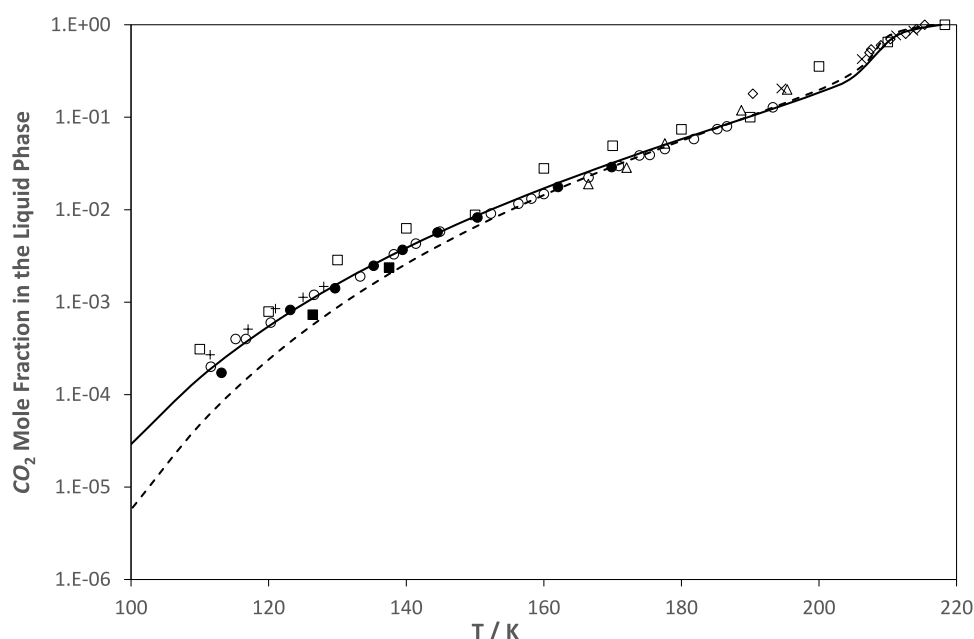


Figure 11. Solubility limit of carbon dioxide in the liquid phase of methane + carbon dioxide mixture at $p = 7$ MPa. Solid-phase model used in calculation: —: Model A; — —: Model D. Experimental data: +: Boyle;⁴² ×: Brady et al.;⁴³ ◇: Brewer and Kurata;⁴⁴ △: Cheung and Zander;⁴⁵ □: Streich;⁴¹ ○: Voss;⁴⁶ ■: Preston et al.;⁴⁷ ●: Gao et al.⁴⁸

Conversely, despite initially appearing complex, Model B presents itself as a relatively straightforward model to implement. Despite the empirical nature of its functional forms for the Gibbs energy expression, the model achieves satisfactory results for the thermodynamics properties considered in this study. Its primary limitation lies in its inability to fully describe certain properties across a temperature and pressure range as broad as that of Model A, as evidenced by discrepancies in molar volume predictions at high temperatures and pressures, and in isobaric heat capacities at low temperatures.

Model A performs well under extreme conditions but presents notable errors in estimating the isobaric thermal expansion coefficient. As previously discussed, this issue could be addressed by increasing the complexity of the Gibbs energy expression (for example, by incorporating a fifth-order temperature dependency to better capture the thermal effects on the isobaric thermal expansion coefficient). However, given that the main focus of this work is to derive a simple EoS for calculating CO_2 solubility at low temperatures, the aforementioned limitation of the EoS is acknowledged but not further pursued.

3.3. Prediction of CO_2 Solubility in Binary Mixtures. To investigate the impact of the different solid- CO_2 models on the prediction of its solubility in fluid solvents, this section compares the calculations obtained for the carbon dioxide + methane and carbon dioxide + nitrogen systems by coupling the same fluid-phase model with four solid-phase models.

The chemical potential of the pure carbon dioxide in the solid phase was calculated either by Model A (eqs 7 and 8), by Model B (Jager and Span),¹¹ by Model C (Trusler),⁹ or by the Classical Approach (eq 20, hereafter called Model D).

Model D corresponds to a simplified version of the Classical Approach,¹² which only considers the latent heat of fusion, $\Delta H_{\text{fus}} = H^L - H^S = 8875$ kJ/mol,¹¹ for evaluating the ratio between the fugacity of pure carbon dioxide in the solid phase, f^S , and that in its liquid phase, f^L :

$$\ln \frac{f^L(T, p)}{f^S(T, p)} = \frac{\Delta H_{\text{fus}}}{RT_t} \left(\frac{T_t}{T} - 1 \right) \quad (20)$$

The chemical potential of carbon dioxide in the fluid phase was calculated using the default multiparameter EoS as implemented in REFPROP v10.⁴ The REFPROP v10 default mixture model uses the reducing and departure functions of GERG-2004⁵ coupled with specific pure fluid EoSs for the pure contributions, namely:

- The Helmholtz EoS for carbon dioxide of Span and Wagner¹⁰
- The Helmholtz EoS for methane of Setzmann and Wagner³⁹
- The Helmholtz EoS for nitrogen of Span et al.⁴⁰

The solid–fluid equilibrium condition was solved based on the equality between the Gibbs energy of carbon dioxide in the solid phase, and the chemical potential of carbon dioxide in the fluid mixture, as expressed in eq 21:

$$\mu_{\text{CO}_2}^S(T, p) = \mu_{\text{CO}_2}^F(T, P, \bar{x}) \text{ with } F = \text{liquid or vapor} \quad (21)$$

Equation 21 assumes that the solid phase consists exclusively of pure dry ice.

The parameters of Model A and B were regressed such that the Gibbs energies of dry ice and the fluid phase were numerically equal at the reference triple-point pressure and temperature (Table 2). This ensures that eq 21 can be directly applied for equilibrium calculations in Excel using the Microsoft Excel version of REFPROP v10.⁴

In contrast, Trusler⁹ chose to match the Gibbs energies at the triple-point temperature and pressure using the default reference state originally selected by Span and Wagner,¹⁰ which differs from that implemented in REFPROP v10.⁴ As a result, entropy and enthalpy offsets are required to adjust the Gibbs energy of Model C, using eq 22:

$$\mu_{\text{CO}_2}^s(T, p) = g_{\text{CO}_2}^{S, \text{model C}}(T, p) - \Delta h + T\Delta s \quad (22)$$

with $\Delta h = 13344.4 \text{ J} \times \text{mol}^{-1}$ and $\Delta s = 93.140 \text{ J} \times \text{K}^{-1} \times \text{mol}^{-1}$.

The reference state has no impact on Model D. Therefore, the iso-fugacity condition can be directly implemented for equilibrium calculations. This condition requires that the fugacities of carbon dioxide in the solid phase and the fluid mixture are equal:

$$f_{\text{CO}_2}^s(T, p) = f_{\text{CO}_2}^f(T, P, \bar{x}) \quad (23)$$

Figure 11 compares experimental values of CO_2 mole fraction in the liquid phase at solid–liquid equilibrium with predictions obtained at 7 MPa, made by coupling the fluid-phase model⁴ with either Model A or Model D. This pressure is higher than the maximum pressure of the SLVE locus of the methane + carbon dioxide system, thus it is suitable for calculating the SLE behavior of the mixture from the triple-point temperature of CO_2 down to liquefied natural gas temperatures.

Using Model A, the experimental data are well represented, successfully capturing CO_2 solubilities as low as a hundred ppm at around 110 K. Deviations were observed in the data reported by Streich,⁴¹ which suggested higher CO_2 solubility compared to the rest of the experimental data.

In contrast, applying Model D results in notable deviations emerging at temperatures below 150 K. This indicates that Model D provides reliable predictions only within a limited temperature range, near the triple point or normal boiling point conditions.

Predictions obtained using Models B and C are not included in Figure 10. This is because the models yielded results nearly identical to those of Model A, as observed by the deviations presented in Table 5 which compares the modeling results with the experimental data shown in Figure 11.

Table 5. Deviations between Solubilities of CO_2 at Solid–Liquid Equilibrium of the CO_2+CH_4 Mixture Calculated by Means of the Four Models for Solid CO_2 (When Coupled with the Reference EoS of Fluid Mixture⁴) and Experimental Values

model	AAD%	bias%	MAD%
A	16.83	−5.18	50.97
B	16.80	−4.80	51.43
C	16.79	−5.54	51.24
D	29.03	−24.76	84.75

For Models A, B, and C, the AAD is approximately 17%, with the MAD reaching 51% (associated with the data of Streich).⁴¹ All three models exhibit a negative bias, indicating a slight underestimation of the solubility. From Table 5, it is clear that Model D provide the least satisfactory results, demonstrating poorer performance in terms of statistical indices compared to the other three models. This highlights the significant impact of the choice of the solid-phase model on solubility and phase calculations.

In Figure 12, the solubility of CO_2 in liquid nitrogen at three different pressures ($P = 4, 6$, and 9 MPa), as predicted by Model A, is compared to literature values reported by Yakimenko et al.⁴⁹

Predictions obtained with Model A align with the qualitative trend of increasing CO_2 solubility with temperature; however, they consistently overestimate the CO_2 concentrations

compared to experimental data. The same behavior is also observed for Models B and C. This systematic discrepancy may, in part, be attributed to the fluid-phase model used in the calculations. The default multiparametric EoS in REFPROP v10⁴ uses the GERG-2004⁵ mixing rule for the $\text{CO}_2\text{--N}_2$ binary system, with mixing parameters fitted to Vapor–Liquid Equilibrium (VLE) and pressure–density–temperature ($p\rho T$) data over the temperature range of 209 to 673 K and CO_2 mole fractions from 0.1 to 0.99.⁵⁰

As a result, the SLE conditions explored in this work extend beyond the temperature and composition ranges for which the mixing rule was originally validated. This may potentially explain the observed systematic deviations between predicted and experimental solubilities.

Unlike Models A, B, and C, the Classical Approach (Model D) consistently predicts lower solubilities than those observed experimentally. Its AAD% is approximately twice as large as that of the other three models, highlighting once again the limitations of this approach when applied to conditions significantly far from the triple or boiling point of the solid-forming component.

Figure 13 shows the comparison between predicted and experimental solubilities of carbon dioxide in supercritical nitrogen.

As shown in Figure 13 and according to the deviations reported in Table 7, Models A, B, and C exhibit comparable behavior, each providing a similar systematic overestimation of CO_2 solubility when coupled with the same fluid-phase model. Interestingly, although Model D yields the poorest representation of CO_2 solubility in liquid nitrogen and in liquid methane (see Tables 5 and 6) compared to Models A–C, it gives the lowest AAD% among the four models in the representation of CO_2 solubility in supercritical nitrogen. Moreover, it shows a less systematic trend of over- or underestimation in solubility predictions, as indicated by a Bias value closer to zero than those obtained for the other three models (Table 7).

4. CONCLUSIONS

Analysis of the temperature dependency of the heat capacity at low temperatures and molar volume of solid phase I of CO_2 at high pressures revealed that the original functional form used for methane and benzene were not entirely suitable. Consequently, the original functional forms used were modified to better capture the thermophysical behavior of dry ice. Despite these efforts, a single continuous and analytically integrable equation for the solid-phase heat capacity could not be achieved across the whole temperature range. The final model was thus formulated as two separate functional forms, each valid within a different temperature region, separated at the inflection temperature of 16.3 K.

This choice was guided by the intent to maintain analytical and easily integrable expressions for the Gibbs energy EoS, ensuring straightforward implementation of the model. The resulting solid model is valid up to 1.2 GPa and 400 K. The parameters of the new EoS were regressed against a comprehensive collection of available experimental data, covering molar volume, isobaric heat capacity, isothermal compressibility, as well as melting and sublimation pressures.

It was also mentioned in this work that the newly proposed model is slightly less accurate than the other two literature models^{9,11} in reproducing the solid thermal expansion coefficient (α) behavior. Nevertheless, the authors decided not to further modify the functional form for molar volume to preserve simplicity of the Gibbs expressions and focus on the

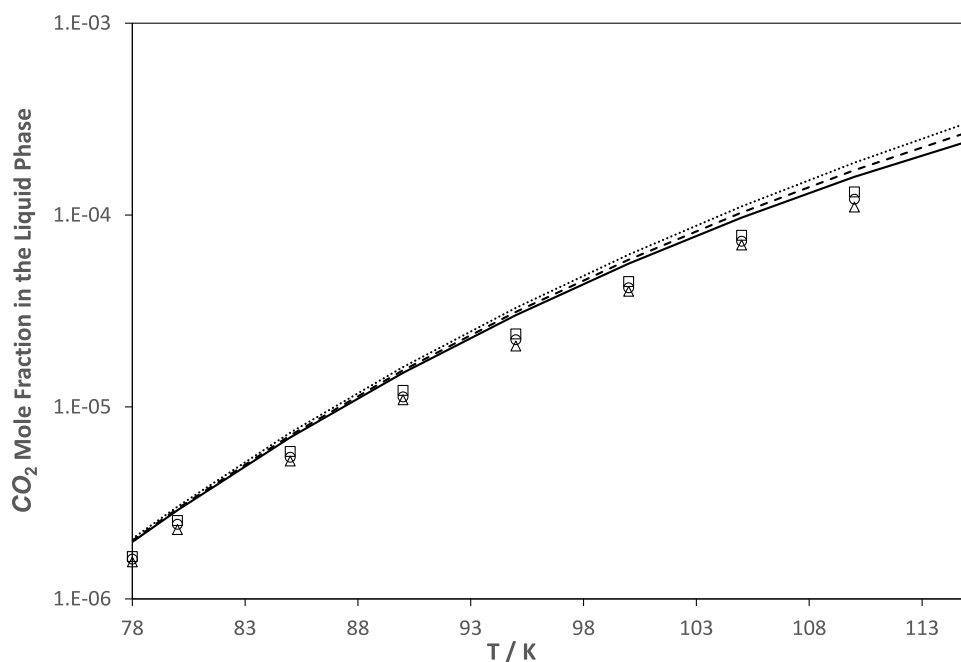


Figure 12. Solubility limit of solid phase I of carbon dioxide in the liquid phase of nitrogen + carbon dioxide mixture at various pressures. Modeling results (Model A): —: 4 MPa; — —: 6 MPa; •••: 9 MPa; Experimental data of Yakimenko et al.:⁴⁹ Δ : 4 MPa; $\circ$: 6 MPa; $\square$: 9 MPa;.

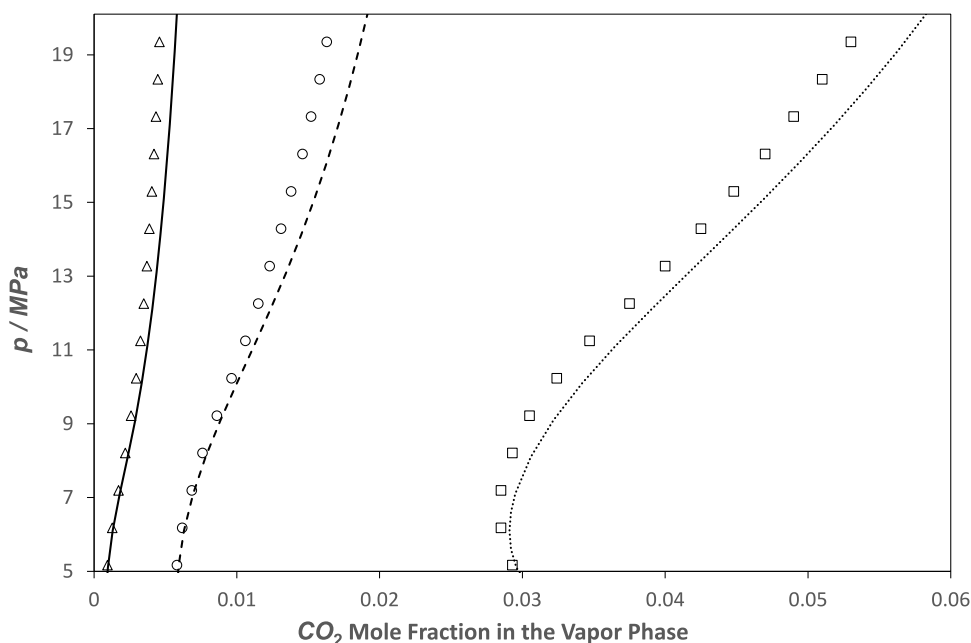


Figure 13. Solubility limit of solid phase I of carbon dioxide in the vapor phase of nitrogen + carbon dioxide mixture at various temperatures. Modeling results (Model A): —: 150 K; — —: 170 K; •••: 190 K; Experimental data of Sonntag et al., 1962⁵¹ and Sonntag et al., 1963:⁵² Δ : 150 K; $\circ$: 170 K; $\square$: 190 K;.

primary goal of predicting solid–fluid equilibria, at the slight detriment of α . Overall, when compared to the literature models,^{9,11} the new Gibbs EoS performs equivalently in representing the other thermodynamic properties for pure CO_2 and solid–fluid phase equilibria of mixtures.

The second part of this work reaffirms that by coupling an accurate fluid-phase EoS with an accurate solid-phase model, reliable predictions of solid–fluid equilibrium can be achieved down to temperatures significantly below the triple-point of the heavy component (e.g., nitrogen/methane + carbon dioxide

mixtures), where the accuracy of the Classical Approach becomes questionable.

Future work will focus on generalizing the functional forms for heat capacity and molar volume, and thus for the Gibbs energy EoS, for other chemicals of interest, including main air components and hydrocarbons, provided that sufficient literature data exist. Additionally, the modeling approach will be applied to the evaluation of the solid–fluid equilibrium behavior of multicomponent mixtures involving one or more heavy components – each assumed to solidify in its pure solid

Table 6. Deviations between Solubilities of CO₂ at Solid–Liquid Equilibrium of the CO₂+N₂ Mixture Calculated by Means of the Four Models for Solid CO₂ (When Coupled with the Reference EoS of Fluid Mixture⁴) and Experimental Values

model	AAD%	bias%	MAD%
A	35.15	35.15	45.39
B	37.55	37.55	47.23
C	34.23	34.23	44.58
D	76.32	−76.32	95.64

Table 7. Deviations between Solubilities of CO₂ at Solid–Vapor Equilibrium of the CO₂+N₂ Mixture Calculated by Means of the Four Models for Solid CO₂ (When Coupled with the Reference EoS of Fluid Mixture⁴) and Experimental Values

model	AAD%	bias%	MAD%
A	9.45	9.40	24.84
B	9.68	9.64	25.33
C	9.06	8.96	24.15
D	8.43	−1.67	23.60

state – to evaluate and compare its predictive capabilities with respect to experimental data for more complex systems.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jced.5c00260>.

First and second partial derivatives of c_p^s with respect to temperature, relationships between the parameters of g^s and parameters of c_p^s , partial derivatives of g^s , statistical indexes (PDF)

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Notes

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